Junbo (Timothy) Du

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Education

University of Illinois Urbana-Champaign — Champaign, Illinois
Computer Science, PhD — *Sept. 2026 – April 2031*

University of Toronto — Toronto, Ontario
Bachelor of Science, Computer Science (3.89 / 4.00 GPA) — *Sept. 2021 – April 2026*

Publications and Awards

Publications:

Sound Source Localization for Human-Robot Interaction in Outdoor Environments
Timothy Du, Victor Liu, Jordy Sehn, Jack Collier, François Grondin.
Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems, **IROS 2025**

Awards:

- Honorable Mention 2025-2026 CRA Outstanding Undergraduate Researcher Award.
- Dean's List Winter 2023, 2024 (*University of Toronto*)
- St. Michael's College Admission Scholarship 2021 (*University of Toronto*)

Research

Hyperscale Data Management — May 2024 – August 2026
Far Data Lab - University of Toronto — *Ontario, CAN*

- Lead the development of SwanLake, a distributed shuffle framework that leverages the growing feasibility of in-memory computing for large scale data analytics and cloud workloads
- Analyzed the push based execution model in DuckDB to identify pipeline breakers, informing the design of a system agnostic shuffle operator that integrates readily into diverse OLAP query engines
- Architected an RDMA network ring buffer that eliminates memory copies and message fragmentation through receiver side pointer indirection and two level commit pointers
- Implemented the coordinator and worker front ends of the system, including zero copy data loading with Apache Arrow and a custom SQL parser enabling query post-processing and DDL support

KidneyOS — May 2024 – Dec 2024
University of Toronto — *Ontario, CAN*

- Led development of the memory allocation subsystem for KidneyOS, a 32-bit operating system written in Rust that enables instructors to teach core OS concepts such as paging and scheduling through hands-on assignments
- Built a frame allocator with an accompanying core map to enable paging support and mitigate internal and external fragmentation in main memory
- Engineered a fixed size sub block allocator achieving amortized constant time allocations by embedding metadata in unused pages to serve dynamically sized kernel memory requests

Professional Experience

Robotics Research Assistant — Sept 2024 – May 2025
Defense Research and Development Canada (DRDC) — *Alberta, CAN*

- Designed a sound source localization (SSL) pipeline for human robot interaction, achieving model accuracy above 95% in both noise relaxed and noise dominant environments
- Optimized the Acoustic Echo Cancellation (AEC) and SRP-PHAT stages of the SSL pipeline by rewriting them in C++ with OpenMP, reducing model latency and delivering a 250% overall speedup
- Initiated the migration of teleoperation and autonomous control ROS nodes on an unmanned ground vehicle to ROS2, and created a software suite automating environment and dependency setup to accelerate development

Technical Skills

Languages: Python, C/C++, Rust, Java, SQL, JavaScript, HTML/CSS, R, Scala
Frameworks & Libraries: PyTorch, Hugging Face, NumPy, Pandas, Apache Arrow, UCX, OpenMP, ROS/ROS2
Developer Tools: Git, Docker, Pip, CMake, Cargo